

Robot That Can Screw In Lightbulb

Northeastern University (Boston, USA) researchers have developed a hybrid robot that blends the torque of rigid robots with the flexibility of soft ones. Inspired by elephant trunks and car CV joints, this robot arm is sensitive and strong.



Hybrid soft and hard robot (Credit: Matthew Modoono/Northeastern University)

Unlike traditional robots, which are either powerful but dangerous, or safe but weak, this new design uses a novel, flexible material to apply precise force safely. The result: a robot that can screw in a lightbulb—safely and efficiently.

Artificial Nerve With Transistors For Brain-Machine Links

Researchers from Xi'an Jiaotong University and the Technical University of Munich have developed a high-frequency artificial nerve using homogeneously integrated vertical N-type and P-type organic electrochemical transistors. This system overcomes ionic/electronic transport and memory retention limitations, delivering fast response and long-term ion storage. It mimics biological nerve components—receptors, synapses, and somas—and functions in sensing, processing, and memory. Successfully tested in mice with neural impairments, it demonstrated biocompatibility and conditioned reflex behaviour. This technology holds promise for neural repair, prosthetics, brain-computer interfaces, and advanced neuro-monitoring systems.

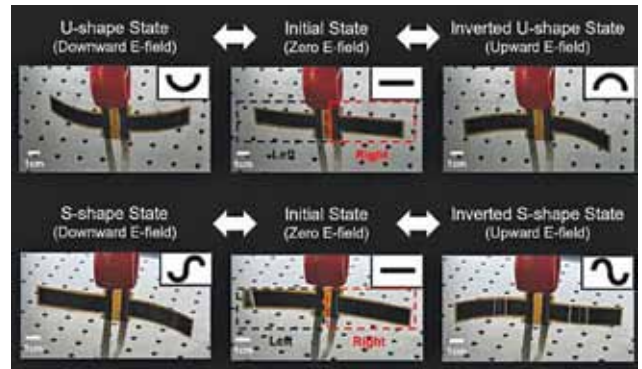
Smarter, Lighter E-Skin With Magnetoreception

Researchers at Helmholtz-Zentrum Dresden-Rossendorf (HZDR) (Dresden, Germany) have created an ultra-thin, transparent e-skin that detects magnetic fields using a single global sensor. This lightweight, flexible membrane simplifies e-skin technology by replacing multiple sensors with a magnetosensitive layer. Using tomography to map magnetic signals, the innovation enables touchless control,

wearable tech advancements, and enhanced robotics. Its simplicity, efficiency, and resistance to interference mark a significant leap in human-machine interaction and smart device integration.

Shape-Changing OLED Panel That Works As A Speaker

Researchers at POSTECH (South Korea) have developed the world's first smartphone-sized OLED panel that changes shape and functions as a speaker, without sacrificing its thin, flexible design. This uses an ultra-thin piezoelectric



Multi-shape deformation without mechanical hinge system (Credit: POSTECH)

polymer actuator integrated with the OLED panel to enable dynamic shape transformations (e.g., concave, S-shaped) via electrical signals—no hinges or motors needed. The actuator also emits high-frequency vibrations, turning the screen into a speaker. Successfully demonstrated on real OLED panels, this tech merges audio output and shape adaptability, opening doors for next-gen foldable devices, wearables, automotive displays, and soft robotics.

Programmable Pixels Transform Infrared Technology

Researchers at Carnegie Mellon (Pennsylvania, United States) have developed an infrared technology: the electrically programmable graphene field-effect transistor (Gr-FET). This active metasurface uses gold pixel arrays and graphene layers to precisely control mid-infrared light across wavelengths, directions, and polarisations. Key applications include preventing thermal side-channel attacks in cybersecurity and enabling early cancer detection by monitoring temperature changes in wearable devices. The device supports scalable 2D pixel control and integrates with existing systems. Promising widespread adoption within 5-10 years, this innovation could transform autonomous vehicles, AR, medical imaging, and next-gen security systems.